

Olist E-Commerce Data Warehouse

Business Analysis — Findings & Answers

Prepared by: Varun

Stack: PostgreSQL (Medallion Architecture — Bronze / Silver / Gold) + Power BI

Scope: 44 business questions across Sales, Customers, Sellers, Products, Delivery, Payments, Reviews, and Geography

Note: All figures are calculated from the silver layer, using orders with order_status = 'delivered' unless otherwise stated. Currency values are in the dataset's native units (BRL).

■ Sales & Revenue

Q1. What is the total revenue and total number of orders overall?

Metric	Value
Total revenue — all order statuses	16,008,873.00
Total revenue — delivered orders only	15,422,462.00
Total orders (all statuses)	99,441

Q2. What is the month-over-month revenue trend across the full dataset?

Year-Month	Month	Total Revenue
2017-01	January 2017	127,545.67
2017-02	February 2017	271,298.65
2017-03	March 2017	414,369.39
2017-04	April 2017	390,952.18
2017-05	May 2017	567,066.73
2017-06	June 2017	490,225.60
2017-07	July 2017	566,403.93
2017-08	August 2017	646,000.61
2017-09	September 2017	701,169.99
2017-10	October 2017	751,140.27
2017-11	November 2017	1,153,528.05
2017-12	December 2017	843,199.17
2018-01	January 2018	1,078,606.86
2018-02	February 2018	966,510.88
2018-03	March 2018	1,120,678.00
2018-04	April 2018	1,132,933.95
2018-05	May 2018	1,128,836.69
2018-06	June 2018	1,012,090.68
2018-07	July 2018	1,027,903.86
2018-08	August 2018	985,414.28

Q3. What is the month-over-month revenue growth rate (%)?

Month	Current Rev.	Prev. Rev.	Growth %
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Feb 2017	271,298.65	127,545.67	+112.71
Mar 2017	414,369.39	271,298.65	+52.74
Apr 2017	390,952.18	414,369.39	-5.65
May 2017	567,066.73	390,952.18	+45.05
Jun 2017	490,225.60	567,066.73	-13.55
Jul 2017	566,403.93	490,225.60	+15.54
Aug 2017	646,000.61	566,403.93	+14.05
Sep 2017	701,169.99	646,000.61	+8.54
Oct 2017	751,140.27	701,169.99	+7.13
Month	Current Rev.	Prev. Rev.	Growth %
Nov 2017	1,153,528.05	751,140.27	+53.57
Dec 2017	843,199.17	1,153,528.05	-26.90
Jan 2018	1,078,606.86	843,199.17	+27.92
Feb 2018	966,510.88	1,078,606.86	-10.39
Mar 2018	1,120,678.00	966,510.88	+15.95
Apr 2018	1,132,933.95	1,120,678.00	+1.09
May 2018	1,128,836.69	1,132,933.95	-0.36
Jun 2018	1,012,090.68	1,128,836.69	-10.34
Jul 2018	1,027,903.86	1,012,090.68	+1.56
Aug 2018	985,414.28	1,027,903.86	-4.13

Q4. Which product category generates the highest total revenue?

beleza_saude (Health & Beauty) — Revenue: **1,412,089.53**

Q5. Which product category has the highest order volume?

cama_mesa_banho (Bed, Bath & Table) — Total items sold: **10,953**

Q6. What is the average order value (AOV) overall, and how does it vary by state?

Overall AOV: **159.86**

State	Orders	AOV
PB	517	266.60
AC	80	244.83
AP	67	240.92

AL	397	237.27
RO	243	234.47
PA	946	224.13
PI	476	221.16
RR	41	220.48
TO	274	219.01
RN	474	212.51
SE	335	209.82
CE	1,279	208.34
MA	717	206.15
MT	886	204.79
PE	1,593	194.02
MS	701	191.76
AM	145	190.32
BA	3,256	181.59
GO	1,957	170.82
SC	3,546	167.85
RJ	12,350	166.45
State	Orders	AOV
DF	2,080	166.42
RS	5,345	161.24
MG	11,354	160.23
ES	1,995	159.24
PR	4,923	158.83
SP	40,500	142.48

Q7. What is the average number of items per order?

Average items per order: **1.14**

Q8. What is the busiest month/season for orders?

By calendar month name (combining both years), **August** is busiest with **10,544** total orders. Looking at individual year-months specifically, **November 2017** alone was the single busiest month with **7,544** orders — likely tied to Black Friday seasonal demand.

Q9. What day of the week sees the most orders placed?

Day	Orders
Monday	15,701
Tuesday	15,503
Wednesday	15,076
Thursday	14,323
Friday	13,685
Sunday	11,635
Saturday	10,555

■ Customers

Q10. How many unique customers are there vs. total orders placed?

Metric	Value
Total unique customers	96,096
Total orders	99,441

Q11. Who are the repeat customers (more than one order)?

Total customers with repeated orders: **2,997**

Q12. What % of total revenue comes from repeat vs. one-time customers?

Customer Type	Total Revenue	Share (%)
One-Time Buyers	14,558,104.56	94.40
Repeat Purchasers	864,357.21	5.60

Q13. What is the average time gap between a customer's first and second order?

Average time gap: **81.20 days**

State	City	Total Customers
SP	Sao Paulo	15,540
RJ	Rio De Janeiro	6,882
MG	Belo Horizonte	2,773

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DF	Brasilia	2,131
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What

customer retention look like by monthly cohort?

Full result set contains **219 rows** (cohort month × month gap). Sample below:

Cohort Month	Month Gap	Retained Customers
2016-09-01	0	1
2016-10-01	0	262
2016-10-01	6	1
2016-10-01	9	1
2016-10-01	11	1

Q15. Which states/cities have the most customers?

Top 5 states:

State	Total Customers
SP	41,746
RJ	12,852
MG	11,635
RS	5,466
PR	5,045

Top 5 cities:

State	City	Total Customers
PR	Curitiba	1,521

Q16. Which states/cities generate the highest revenue per customer?

Top 5 states:

State	Revenue/Customer	Total Customers
PB	273.48	504
AC	257.71	76
RO	246.65	231
AP	244.57	66
AL	243.40	387

Top 5 cities (note: driven by single high-value orders in low-volume cities):

State	City	Revenue/Customer	Customers
PB	Pianco	2,324.99	1
AL	Sao Sebastiao	2,269.98	1

PA	Nova Esperanca Do Piria	2,252.66	1
MG	Engenheiro Navarro	2,106.55	1
PE	Agrestina	2,066.34	1

■ Sellers

Note: seller revenue figures below use *price only* (excludes freight_value, which is a pass-through logistics cost, not seller earnings).

Q17. Who are the top 5 sellers by total revenue?

Seller ID	Total Revenue
4869f7a5dfa277a7dca6462dcf3b52b2	226,987.93
53243585a1d6dc2643021fd1853d8905	217,940.44
4a3ca9315b744ce9f8e937436149388	196,882.12
fa1c13f2614d7b5c4749cbc52fecda94	190,917.14
7c67e1448b00f6e969d365cea6b010ab	186,570.05

Q18. How concentrated is revenue among the top 10% of sellers?

Metric	Value
Top 10% seller revenue	8,873,076.37
Total revenue (all sellers)	13,221,498.11
Revenue share of top 10%	67.11%

Q19. How does seller count and performance vary by state?

State	Sellers	Revenue	Avg Rev/Seller	Avg Rating	On-Time %
SP	1,849	8,548,955.78	4,623.56	4.06	91.28
PR	349	1,240,516.38	3,554.49	4.12	93.55
MG	244	981,274.22	4,021.62	4.17	94.36
RJ	171	823,390.38	4,815.15	4.15	91.63
SC	190	615,030.19	3,237.00	4.13	93.99

Q20. What is the average revenue per seller?

Average revenue per seller: **4,451.68**

Q21. Which sellers have the most orders but lowest average review score?

Seller ID	Total Orders	Avg Rating
1ca7077d890b907f89be8c954a02686a	108	2.39
2eb70248d66e0e3ef83659f71b244378	187	2.79
a49928bcdf77c55c6d6e05e09a9b4ca5	96	3.03
972d0f9cf61b499a4812cf0bfa3ad3c4	81	3.03
54965bbe3e4f07ae045b90b0b8541f52	73	3.14

2a1348e9addc1af5aaa619b1a3679d6b	48	3.17
bbad7e518d7af88a0897397ffdca1979	68	3.18
5058e8c1e82653974541e83690655b4a	62	3.25
602044f2c16190c2c6e45eb35c2e21cb	48	3.32
ad781527c93d00d89a11eecd9dcad7c1	38	3.34



Q22. Which are the top 10 best-selling categories by revenue?

Category	Revenue
health_beauty	1,233,131.72
watches_gifts	1,166,176.98
bed_bath_table	1,023,434.76
sports_leisure	954,852.55
computers_accessories	888,724.61
furniture_decor	711,927.69
housewares	615,628.69
cool_stuff	610,204.10
auto	578,966.65
toys	471,286.48

Q23. Which are the top 10 best-selling categories by quantity sold?

Category	Items Sold
bed_bath_table	10,953
health_beauty	9,465
sports_leisure	8,431
furniture_decor	8,160
computers_accessories	7,644
housewares	6,795
watches_gifts	5,859
telephony	4,430
garden_tools	4,268
auto	4,140

Q24. Which categories have the highest and lowest average review scores?

Tier	Category	Reviews	Avg Rating
Highest	books_general_interest	533	4.51

Highest	books_imported	57	4.51
Highest	small_appliances_home_oven	73	4.44
Highest	construction_tools_tools	99	4.44
Highest	flowers	31	4.42
Lowest	audio	359	3.83
Lowest	fixed_telephony	253	3.76
Lowest	fashion_male_clothing	124	3.76
Lowest	office_furniture	1,664	3.52
Lowest	diapers_and_hygiene	37	3.38

Q25. Are there categories with high revenue but low review scores?

Category	Items Sold	Revenue	Avg Rating
office_furniture	1,668	268,154.31	3.52
bed_bath_table	10,953	1,023,434.76	3.93
furniture_decor	8,160	711,927.69	3.96
computers_accessories	7,644	888,724.61	3.99
telephony	4,430	309,860.23	3.99

Q26. What is the average price per product category?

Category	Avg Price	Items Sold
computers	1,098.92	199
small_appliances_home_oven	638.21	73
home_appliances_2	467.33	231
agro_industry_and_commerce	342.55	206
musical_instruments	283.13	651

Q27. Which categories have the widest price range?

Category	Price Range
housewares	6,731.94
computers	6,694.50
art	6,495.50
small_appliances	4,792.50
musical_instruments	4,394.97



Q28. What is the average delivery time (purchase to delivery) overall?

Average delivery time: **12.56 days**

Q29. How does average delivery time vary by state?

Top 5 states with longest delivery times:

State	Avg Delivery Time (days)
RR	29.39
AP	27.19
AM	26.43
AL	24.54
PA	23.77

Q30. What percentage of orders are delivered late?

Late delivery rate: **8.11%**

Q31. Which states experience the most delivery delays?

State	Total Orders	Late Deliveries
SP	40,480	2,387
RJ	12,349	1,664
MG	11,349	637
BA	3,256	457
RS	5,343	382

Q32. Is there a relationship between delivery delay and review score?

Yes — late deliveries are clearly associated with lower review scores:

Delivery Performance	Total Orders	Avg Review Score
On Time or Early	88,140	4.30
Late	7,661	2.57

Q33. Average time between order approval and carrier handoff?

Average: **2.80 days**

Q34. Average time between carrier handoff and customer delivery?

Average: **9.33 days**



Q35. What are the most common payment methods used?

Payment Type	Total Payments
credit_card	76,795
boleto	19,784
voucher	5,775
debit_card	1,529
not_defined	3

Q36. Does payment type affect average order value?

Installments	Total Order Value
1	5,907,421.99
10	2,211,577.34
2	1,579,283.03
3	1,491,103.80
8	1,313,423.34



Yes —

4	1,163,907.61	AOV
5	961,174.30	
6	822,611.81	

increases with more installments (credit card), and boleto/debit transactions cluster around a fixed single-payment value. Selected rows:

Type	Installments	Volume	AOV	Total Revenue
boleto	1	19,784	145.03	2,869,361.27
credit_card	1	25,457	95.87	2,440,634.06
credit_card	10	5,328	415.09	2,211,577.34
debit_card	1	1,529	142.57	217,989.79
voucher	1	5,775	65.70	379,436.87

Q37. How often are orders paid in installments?

Metric	Value
Total orders	99,440
Orders using installments	51,170
% using installments	51.46%
Avg installments (all orders)	2.93
Avg installments (when split)	4.75

Q38. Is there a relationship between installment count and order value?

Q39. What is the overall distribution of review scores (1-5)?

Score	Total Reviews	Share (%)
5	57,328	57.78
4	19,142	19.29
3	8,179	8.24
2	3,151	3.18



1	11,424	11.51
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Q40. What percentage of orders receive a review at all?

Orders with Reviews	Total Delivered	Response Rate
95,832	96,478	99.33%

Q41. Is there a correlation between delivery speed and review score?

Delivery Window	Orders	Avg Score
0-5 Days (Fast)	13,416	4.45
6-10 Days (Acceptable)	33,035	4.36
11-15 Days (Slow)	23,606	4.27
16-20 Days (Very Slow)	12,159	4.13
20+ Days (Critical)	14,114	3.23

Q42. Which product categories receive the most negative reviews (score 1-2)?

Category	Negative Reviews
bed_bath_table	2,055
furniture_decor	1,581
computers_accessories	1,435
health_beauty	1,312
sports_leisure	1,251
housewares	1,079
watches_gifts	965
telephony	761
garden_tools	689
auto	637

Q43. Which states have the highest concentration of customers and sellers?

State	Customers	Sellers
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SP	41,746	1,849
RJ	12,852	171
MG	11,635	244
RS	5,466	129
PR	5,045	349
SC	3,637	190
BA	3,380	19
DF	2,140	30
ES	2,033	23
GO	2,020	40

Q44. Is there a pattern between customer-seller distance and delivery time?

Route	Orders	Avg Delivery (days)
Same State (Intrastate)	34,806	7.93
Cross-State (Interstate)	61,885	15.05

Cross-state deliveries take almost **2x longer** on average than same-state deliveries — a clear, expected logistics distance effect.
